

Lab 4: Limited Dependent Variable Models

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1 Goals

1. Fit and read logit and probit models, and see how little the choice matters.
2. Interpret on the probability scale: average marginal effects and predicted probabilities, with intervals.
3. Handle interactions correctly in a nonlinear model.
4. Diagnose and repair separation.
5. Fit count models, detect overdispersion, and choose a remedy.
6. Fit an ordered logit and test proportional odds.
7. Handle censored durations: Kaplan–Meier, the Cox model, and the discrete-time person-period logit (Thursday’s event-history session).

```
library(survival) # Surv, survfit, coxph, survSplit
library(dplyr)
library(ggplot2)
library(marginaleffects)
library(sandwich); library(lmtest)
library(MASS)      # glm.nb, polr
library(AER)       # dispersiontest
library(brglm2)    # Firth penalized likelihood
library(pROC)

gss <- readRDS("../Data/gss_earnings.rds")
```

2 Binary outcomes

Outcome: `ba`, whether the respondent holds a bachelor’s degree or more.

```
f <- ba ~ pareduc + wordsum + female + black + exper

m_lpm    <- lm(f, data = gss)
m_logit  <- glm(f, data = gss, family = binomial(link = "logit"))
m_probit <- glm(f, data = gss, family = binomial(link = "probit"))

round(cbind(LPM = coef(m_lpm),
            logit = coef(m_logit),
            probit = coef(m_probit),
            ratio = coef(m_logit) / coef(m_probit)), 4)

#>               LPM    logit  probit ratio
#> (Intercept) -0.3241 -4.8700 -2.7988 1.740
#> pareduc      0.0297  0.1798  0.1014 1.774
#> wordsum      0.0740  0.4125  0.2448 1.685
#> female       0.0478  0.2780  0.1581 1.759
#> black       -0.0421 -0.1510 -0.0880 1.715
```

```
#> exper      -0.0061 -0.0313 -0.0190 1.647
```

The logit-to-probit ratio hovers around 1.7, as the theory predicts. The two are *different* models, different link functions, different assumed error distributions, but they are close enough empirically that the choice between them almost never matters for the quantities we report.

```
## how often does the LPM predict outside [0,1]?
mean(fitted(m_lpm) < 0 | fitted(m_lpm) > 1)
#> [1] 0.04645
range(fitted(m_lpm))
#> [1] -0.5137 1.0268
```

i Note

The linear probability model is not automatically wrong, but check this number. If a substantial share of fitted values leaves $[0, 1]$, the linear approximation is being asked to do something it cannot.

2.1 Odds ratios

```
round(exp(cbind(OR = coef(m_logit), confint.default(m_logit))), 3)
#>
#> OR 2.5 % 97.5 %
#> (Intercept) 0.008 0.005 0.013
#> pareduc 1.197 1.165 1.230
#> wordsum 1.511 1.440 1.585
#> female 1.320 1.132 1.540
#> black 0.860 0.692 1.069
#> exper 0.969 0.962 0.976
```

Read the **wordsum** row as: each additional correct word multiplies the *odds* of holding a BA by about 1.5. Whether that corresponds to a big or small change in the *probability* depends entirely on where you start, which is why we now move to the probability scale.

2.2 Average marginal effects

```
avg_slopes(m_logit)
#>
#> Term Contrast Estimate Std. Error z Pr(>|z|) S 2.5 % 97.5 %
#> black 1 - 0 -0.02824 0.020682 -1.37 0.172 2.5 -0.06877 0.01230
#> exper dY/dX -0.00588 0.000663 -8.88 <0.001 60.3 -0.00718 -0.00458
#> female 1 - 0 0.05224 0.014702 3.55 <0.001 11.4 0.02343 0.08105
#> pareduc dY/dX 0.03375 0.002342 14.41 <0.001 154.0 0.02916 0.03834
#> wordsum dY/dX 0.07743 0.003912 19.79 <0.001 287.3 0.06976 0.08510
```

```
#>
#> Type: response
```

Compare with the marginal effect at the mean, and with the LPM coefficients:

```
mem <- slopes(m_logit, newdata = "mean")
comparison <- data.frame(
  variable = c("pareduc", "wordsum", "female", "black", "exper"),
  AME = avg_slopes(m_logit)$estimate,
  MEM = mem$estimate,
  LPM = coef(m_lpm)[-1]
)
round(comparison[, -1], 4) |> `rownames<-`(comparison$variable)
#>           AME      MEM      LPM
#> pareduc -0.0282 -0.0316  0.0297
#> wordsum -0.0059 -0.0068  0.0740
#> female  0.0522  0.0628  0.0478
#> black   0.0338  0.0388 -0.0421
#> exper   0.0774  0.0890 -0.0061
```

Three observations worth taking seriously:

- The AME and the LPM coefficients are very close. This is typical, and it is the main reason people defend the LPM for descriptive and causal work.
- The MEM differs from the AME. It answers a different question: the effect for one hypothetical person at the mean of everything, including means of dummy variables.
- Following Hanmer and Kalkan (2013), report the **AME**.

2.3 Predicted probabilities, the observed-value way

```
avg_predictions(m_logit, variables = list(wordsum = c(3, 6, 9)))
#>
#> wordsum Estimate Std. Error    z Pr(>|z|)      S 2.5 % 97.5 %
#>      3    0.159    0.01209 13.1  <0.001 128.6 0.135 0.183
#>      6    0.366    0.00855 42.8  <0.001   Inf 0.349 0.383
#>      9    0.631    0.01467 43.0  <0.001   Inf 0.602 0.659
#>
#> Type: response
```

This sets `wordsum` to each value for *every* respondent, leaving their other covariates alone, computes predicted probabilities, and averages. Contrast with setting everything else to its mean:

```
predictions(m_logit, newdata = datagrid(wordsum = c(3, 6, 9)))
#>
```

```
#> wordsum Estimate Pr(>|z|)      S 2.5 % 97.5 %
#>      3    0.118  <0.001 256.7 0.0974 0.141
#>      6    0.315  <0.001 124.8 0.2900 0.341
#>      9    0.613  <0.001  26.4 0.5749 0.650
#>
#> Type: invlink(link)
```

```
plot_predictions(m_logit, by = "wordsum") +
  labs(x = "Vocabulary score (0-10)", y = "Pr(bachelor's or more)") +
  theme_minimal(base_size = 10)
```

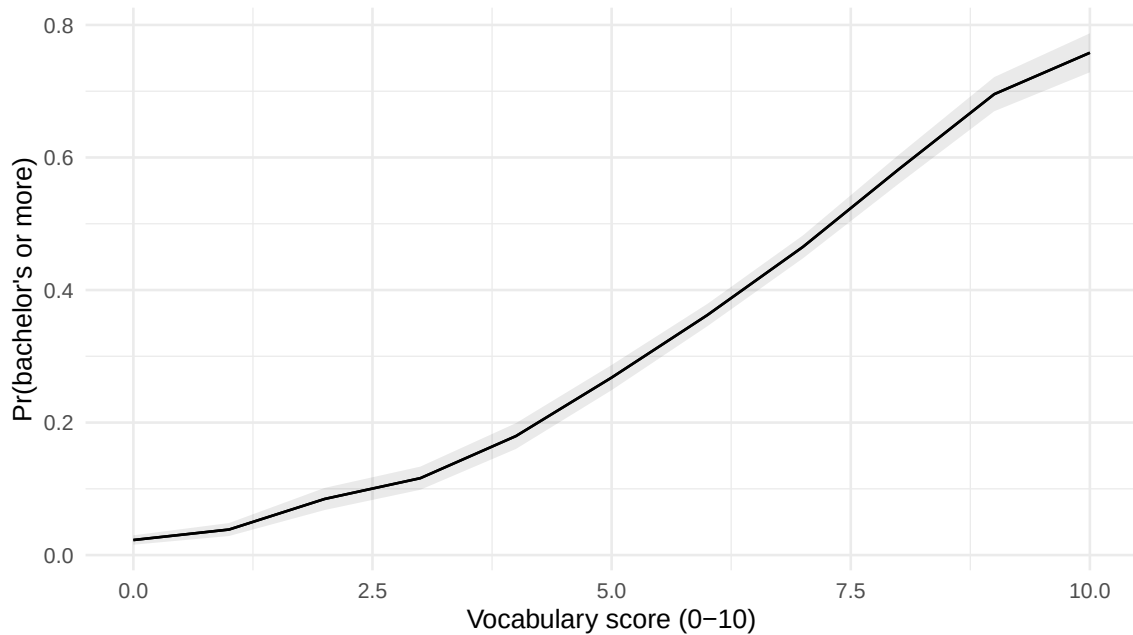


Figure 1: Average predicted probability of holding a BA by vocabulary score (observed-value approach), with 95% intervals.

And the contrast we probably want to report:

```
avg_comparisons(m_logit, variables = list(wordsum = c(3, 9)))
#>
#> Estimate Std. Error z Pr(>|z|)      S 2.5 % 97.5 %
#>    0.472    0.0236 20  <0.001 292.4 0.425 0.518
#>
#> Term: wordsum
#> Type: response
#> Comparison: 9 - 3
```

2.4 Robust and clustered standard errors

Everything from Week 2 carries over, the sandwich works on `glm` objects.

```
round(cbind(
  classical = sqrt(diag(vcov(m_logit))),
  HCO       = sqrt(diag(vcovHC(m_logit, type = "HCO"))),
  cluster   = sqrt(diag(vcovCL(m_logit, cluster = ~ psu)))
), 4)
#>           classical      HCO cluster
#> (Intercept)    0.2640 0.2842  0.2834
#> pareduc       0.0137 0.0157  0.0159
#> wordsum       0.0246 0.0252  0.0250
#> female       0.0785 0.0781  0.0735
#> black        0.1111 0.1119  0.1110
#> exper        0.0037 0.0036  0.0036
```

```
avg_slopes(m_logit, vcov = ~ psu)
#>
#>      Term Contrast Estimate Std. Error      z Pr(>|z|)      S    2.5 %    97.5 %
#> black      1 - 0 -0.02824  0.020651 -1.37    0.172    2.5 -0.06871  0.01224
#> exper      dY/dX -0.00588  0.000648 -9.07   <0.001   62.9 -0.00715 -0.00461
#> female     1 - 0  0.05224  0.013762  3.80   <0.001   12.7  0.02527  0.07921
#> pareduc    dY/dX  0.03375  0.002705 12.48   <0.001  116.3  0.02845  0.03905
#> wordsum    dY/dX  0.07743  0.004021 19.25   <0.001  272.0  0.06955  0.08531
#>
#> Type: response
```

3 Interactions

Does the vocabulary gradient differ by gender?

```
m_int <- glm(ba ~ pareduc + wordsum * female + black + exper,
  data = gss, family = binomial)
round(coef(summary(m_int))[, 1:2], 4)
#>           Estimate Std. Error
#> (Intercept)  -4.4690      0.2983
#> pareduc      0.1800      0.0137
#> wordsum      0.3533      0.0323
#> female     -0.5842      0.3282
#> black      -0.1390      0.1115
#> exper      -0.0318      0.0037
#> wordsum:female 0.1306      0.0483
```

The product-term coefficient is on the **log-odds** scale. What we actually want is whether the

marginal effect on the probability differs by gender:

```
avg_slopes(m_int, variables = "wordsum", by = "female")
#>
#>   female Estimate Std. Error    z Pr(>|z|)    S 2.5 % 97.5 %
#>     0     0.0661    0.00524 12.6  <0.001 118.8 0.0558 0.0763
#>     1     0.0907    0.00548 16.5  <0.001 201.9 0.0800 0.1015
#>
#> Term: wordsum
#> Type: response
#> Comparison: dY/dX
```

```
avg_comparisons(m_int, variables = "wordsum", by = "female",
                 hypothesis = "b2 - b1 = 0")
#>
#> Hypothesis Estimate Std. Error    z Pr(>|z|)    S 2.5 % 97.5 %
#>   b2-b1=0    0.0241    0.00746 3.22  0.00127 9.6 0.00942 0.0387
#>
#> Type: response
```

! Important

Note the two can disagree in sign or significance (Ai and Norton 2003). In a nonlinear model the interaction on the probability scale depends on where each person sits on the curve, not only on the product term. **Always report the quantity you actually mean,** with an interval.

4 Separation

Construct a variable that perfectly predicts the outcome, and watch the MLE run off to infinity.

```
gss2 <- gss |> mutate(perfect = as.integer(educ >= 18 & ba == 1))
m_sep <- glm(ba ~ pareduc + perfect, data = gss2, family = binomial)
round(coef(summary(m_sep))[, 1:2], 3)
#>           Estimate Std. Error
#> (Intercept)  -3.899      0.207
#> pareduc       0.232      0.015
#> perfect      19.320     271.710
```

The `perfect` coefficient is enormous with an enormous standard error, the classic signature. The fix is Firth's penalized likelihood:

```
m_firth <- glm(ba ~ pareduc + perfect, data = gss2,
               family = binomial, method = "brglmFit")
round(coef(summary(m_firth))[, 1:2], 3)
```

```
#>           Estimate Std. Error
#> (Intercept)  -3.893      0.207
#> pareduc      0.232      0.015
#> perfect      7.803      1.424
```

Finite estimates, usable standard errors, and the variable is retained (Zorn 2005).

5 Count outcomes

Outcome: `childs`, the number of children.

```
m_pois <- glm(childs ~ educ + female + black + exper + evermar,
              data = gss, family = poisson(link = "log"))
round(coef(summary(m_pois))[, 1:2], 4)
#>           Estimate Std. Error
#> (Intercept)  -0.0064      0.0876
#> educ        -0.0455      0.0048
#> female       0.0089      0.0269
#> black        0.3600      0.0349
#> exper        0.0118      0.0013
#> evermar      0.9468      0.0430
```

Coefficients are semi-elasticities: e^β is the multiplicative change in the expected count.

```
round(exp(coef(m_pois)), 3)
#> (Intercept)      educ      female      black      exper      evermar
#>      0.994      0.956      1.009      1.433      1.012      2.578
```

5.1 Is it overdispersed?

```
c(mean = mean(gss$childs), var = var(gss$childs))
#> mean var
#> 1.604 2.109

## a score-type test: it uses only the Poisson fit (the restricted model,
## variance = mean) and asks whether the slope toward variance > mean is zero
dispersiontest(m_pois, trafo = 1)
#>
#> Overdispersion test
#>
#> data: m_pois
#> z = 4, p-value = 0.00004
#> alternative hypothesis: true alpha is greater than 0
#> sample estimates:
```

```
#> alpha
#> 0.1676
```

This is lecture's third ruler. The Poisson model is the restricted one, and the test never fits the negative binomial; it reads the slope of the log-likelihood at the Poisson fit in the direction of extra variance.

```
m_qp <- glm(childs ~ educ + female + black + exper + evermar,
             data = gss, family = quasipoisson)
m_nb <- glm.nb(childs ~ educ + female + black + exper + evermar, data = gss)

round(cbind(
  poisson_se   = sqrt(diag(vcov(m_pois))),
  quasi_se     = sqrt(diag(vcov(m_qp))),
  negbin_se    = sqrt(diag(vcov(m_nb)))[names(coef(m_pois))],
  poisson_robust = sqrt(diag(vcovHC(m_pois, type = "HC0")))
), 4)
#>
#>          poisson_se quasi_se negbin_se poisson_robust
#> (Intercept)    0.0876   0.0939   0.0900         0.1012
#> educ          0.0048   0.0051   0.0049         0.0052
#> female        0.0269   0.0288   0.0277         0.0278
#> black         0.0349   0.0374   0.0361         0.0373
#> exper         0.0013   0.0014   0.0014         0.0013
#> evermar       0.0430   0.0461   0.0437         0.0551
```

Note that the coefficients barely move but the standard errors do. That is the signature of overdispersion: the mean is fine, the variance assumption is not.

```
c(theta = m_nb$theta, se_theta = m_nb$SE.theta)
#>      theta se_theta
#>    31.25    14.91
AIC(m_pois, m_nb)
#>      df    AIC
#> m_pois  6 10778
#> m_nb    7 10775
```

A finite $\hat{\theta}$ well away from infinity, and a much lower AIC, both say the negative binomial is doing real work.

5.2 Does the model reproduce the data?

```
set.seed(1)
n <- nrow(gss)
sim <- bind_rows(
  tibble::tibble(k = gss$childs, source = "observed"),
```

```

tibble::tibble(k = rpois(n, fitted(m_pois)), source = "Poisson"),
tibble::tibble(k = rnbinom(n, mu = fitted(m_nb), size = m_nb$theta),
source = "Neg. binomial")
) |>
count(source, k) |>
group_by(source) |> mutate(p = n / sum(n)) |> ungroup() |> # normalize on FULL n
filter(k <= 8) # then trim for display

ggplot(sim, aes(k, p, fill = source)) +
  geom_col(position = "dodge") +
  scale_fill_manual(values = c("observed" = "grey30", "Poisson" = "navy",
                              "Neg. binomial" = "firebrick")) +
  labs(x = "number of children", y = "proportion", fill = NULL) +
  theme_minimal(base_size = 10)

```

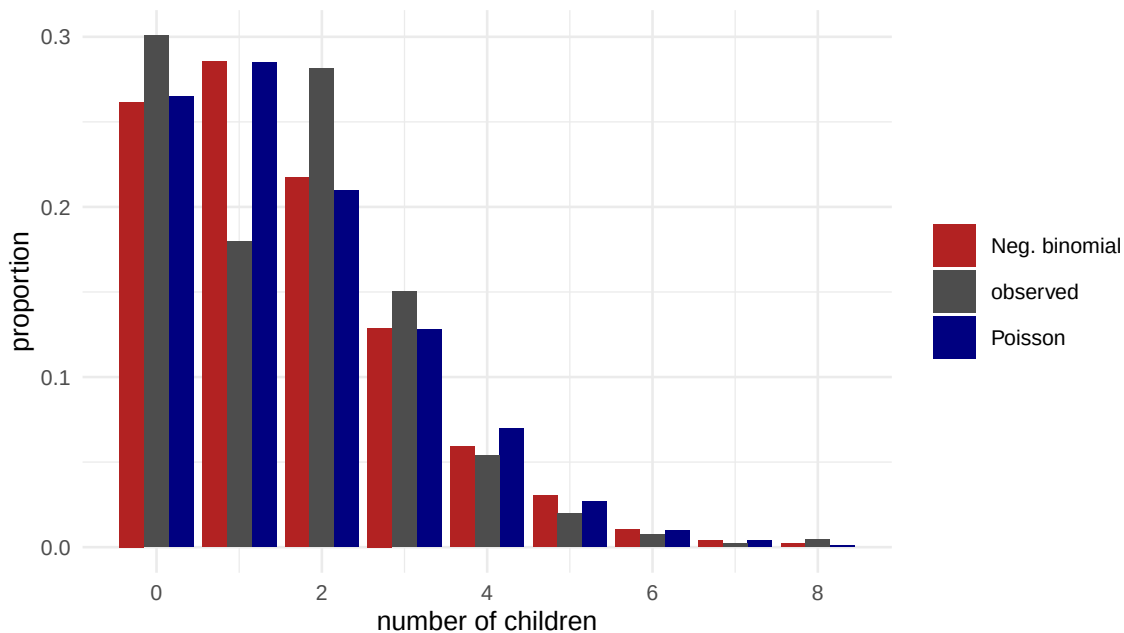


Figure 2: Observed distribution of the number of children against the distributions implied by the Poisson and negative binomial fits.

This *predictive check* is the single most informative count diagnostic, and the one people most often skip.

6 Ordered outcomes

```

m_ord <- polr(degree ~ pareduc + wordsum + female + black,
              data = gss, Hess = TRUE)
round(coef(summary(m_ord)), 4)

```

```
#>
#>      Value Std. Error t value
#> pareduc      0.2082    0.0104 19.9296
#> wordsum      0.3578    0.0192 18.5908
#> female      0.3349    0.0646  5.1829
#> black     -0.0238    0.0914 -0.2607
#> < HS|HS      1.8372    0.1616 11.3667
#> HS|Junior college  5.2081    0.1788 29.1209
#> Junior college|Bachelor 5.6724    0.1823 31.1134
#> Bachelor|Graduate  7.3582    0.1973 37.3032
```

`polr()` does not print p -values; get them from the t -values, or better, work on the probability scale:

```
avg_predictions(m_ord, variables = list(wordsum = c(3, 9)), type = "probs")
#>
#>      Group wordsum Estimate Std. Error      z Pr(>|z|)      S 2.5 % 97.5 %
#> < HS          3    0.1265    0.00888 14.25 <0.001 150.7 0.1091 0.1439
#> < HS          9    0.0190    0.00195  9.72 <0.001  71.8 0.0152 0.0228
#> HS           3    0.6228    0.01180 52.80 <0.001   Inf 0.5997 0.6459
#> HS           9    0.2839    0.01091 26.02 <0.001 493.4 0.2625 0.3053
#> Junior college 3    0.0719    0.00446 16.12 <0.001 191.8 0.0632 0.0807
#> Junior college 9    0.0936    0.00506 18.49 <0.001 251.2 0.0837 0.1035
#> Bachelor      3    0.1373    0.00812 16.90 <0.001 210.5 0.1214 0.1532
#> Bachelor      9    0.3534    0.00993 35.58 <0.001 918.6 0.3339 0.3729
#> Graduate      3    0.0414    0.00367 11.27 <0.001  95.5 0.0342 0.0486
#> Graduate      9    0.2501    0.01092 22.91 <0.001 383.5 0.2287 0.2715
#>
#> Type: probs
```

6.1 Testing proportional odds

The model imposes one β across all cutpoints. Compare it to a multinomial logit, which frees the coefficients:

The proportional-odds model is *nested inside* a generalized ordered logit that lets each cutpoint have its own coefficients, so we can test the restriction by a likelihood-ratio test against that model.

```
library(VGAM)
## proportional odds imposed
m_po <- vglm(degree ~ pareduc + wordsum + female + black,
             family = cumulative(parallel = TRUE), data = gss)
## proportional odds relaxed
m_npo <- vglm(degree ~ pareduc + wordsum + female + black,
```

```

family = cumulative(parallel = FALSE), data = gss)

lr <- 2 * (logLik(m_npo) - logLik(m_po))
df <- length(coef(m_npo)) - length(coef(m_po))
c(LR = lr, df = df, p = pchisq(lr, df, lower.tail = FALSE))
#> LR df p
#> NaN 12 NaN

```

A small p -value says the proportional-odds restriction is rejected. With n this large that is common; look at *how much* the free coefficients differ before abandoning the parsimony. (An ordered logit and a *multinomial* logit are not nested, so their deviances cannot be compared this way.)

7 Model fit

```

AIC(m_logit, m_int)
#>      df AIC
#> m_logit 6 3896
#> m_int    7 3890
BIC(m_logit, m_int)
#>      df BIC
#> m_logit 6 3933
#> m_int    7 3934

## McFadden's pseudo-R2
m_null <- glm(ba ~ 1, data = gss, family = binomial)
1 - as.numeric(logLik(m_logit)) / as.numeric(logLik(m_null))
#> [1] 0.1827

```

```

roc_obj <- roc(gss$ba, fitted(m_logit), quiet = TRUE)
plot(roc_obj, col = "navy", lwd = 2, legacy.axes = TRUE)
text(0.6, 0.2, paste("AUC =", round(auc(roc_obj), 3)))

```

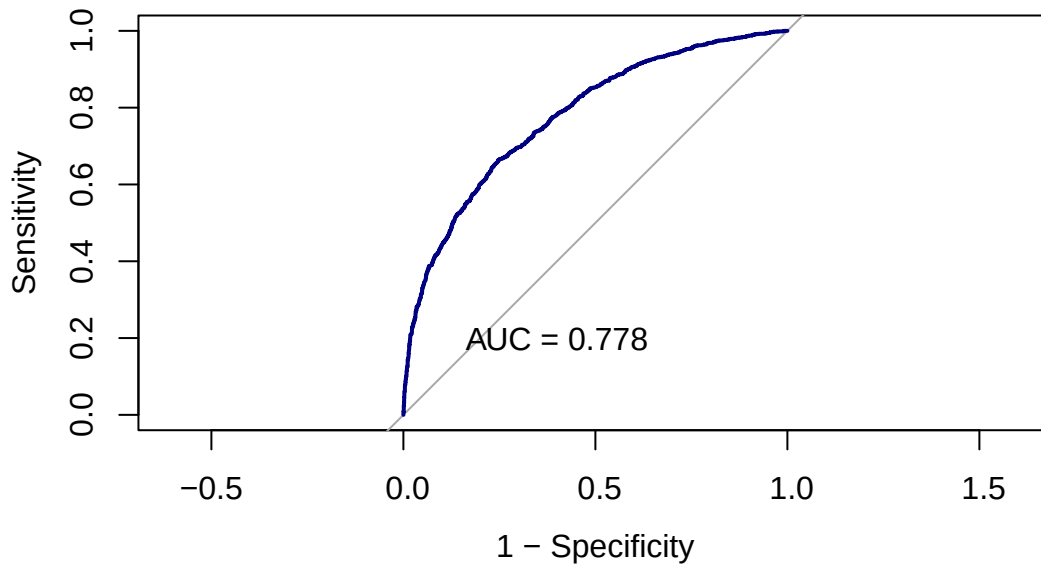


Figure 3: ROC curve for the logit model of holding a bachelor's degree.

```
## classification accuracy against the null rule of "always predict the majority"
pred <- as.integer(fitted(m_logit) > 0.5)
c(model = mean(pred == gss$ba),
  always_majority = max(mean(gss$ba), 1 - mean(gss$ba)))
#>           model always_majority
#>           0.7170           0.5893
```

8 Event history (Thursday's session)

The GSS extract has no duration variable, so we simulate one — which has the usual advantage that we *know the truth* and can see whether each method recovers it.

8.1 A duration with censoring, by construction

Time (in years after age 18) to first union formation. College graduates form unions at a lower rate — the true log hazard ratio is -0.5 — and everyone is observed for at most 12 years, so long durations are censored.

```
set.seed(4)
n <- 3000
sim <- tibble::tibble(
  college = rbinom(n, 1, 0.35),
  rate     = exp(-2.2 - 0.5 * college), # hazard, events per year
  t_true   = rexp(n, rate),             # latent duration
  time     = pmin(t_true, 12),          # observed: censored at 12 years
  event    = as.integer(t_true <= 12)   # 1 = union observed, 0 = censored
)
```

```
sim |> count(college, event)
#> # A tibble: 4 x 3
#>   college event     n
#>   <int> <int> <int>
#> 1     0     0  511
#> 2     0     1 1462
#> 3     1     0  450
#> 4     1     1  577
```

A third to a half of each group is censored. That is the normal situation in event-history data, not a pathology.

8.2 Why the naive fixes fail

Before doing it right, do it wrong twice:

```
sim |>
  group_by(college) |>
  summarise(
    drop_censored = mean(time[event == 1]), # naive 1: drop censored
    censored_as_event = mean(time),          # naive 2: censoring = event
    truth         = mean(1 / rate)           # true expected duration
  ) |> round(2)
#> # A tibble: 2 x 4
#>   college drop_censored censored_as_event truth
#>   <dbl>         <dbl>         <dbl> <dbl>
#> 1     0           4.8           6.66  9.03
#> 2     1           5.35           8.26 14.9
```

Both naive columns understate the true durations, and — worse — they understate them *more for college graduates*, whose long durations are censored more often. The college gap itself is distorted, not just the level.

8.3 Kaplan–Meier

```
km <- survfit(Surv(time, event) ~ college, data = sim)
plot(km, col = c("navy", "firebrick"), lwd = 2,
     xlab = "Years since age 18", ylab = "Share still unpartnered")
legend("bottomleft", c("No college", "College"),
     col = c("navy", "firebrick"), lwd = 2, bty = "n")
```

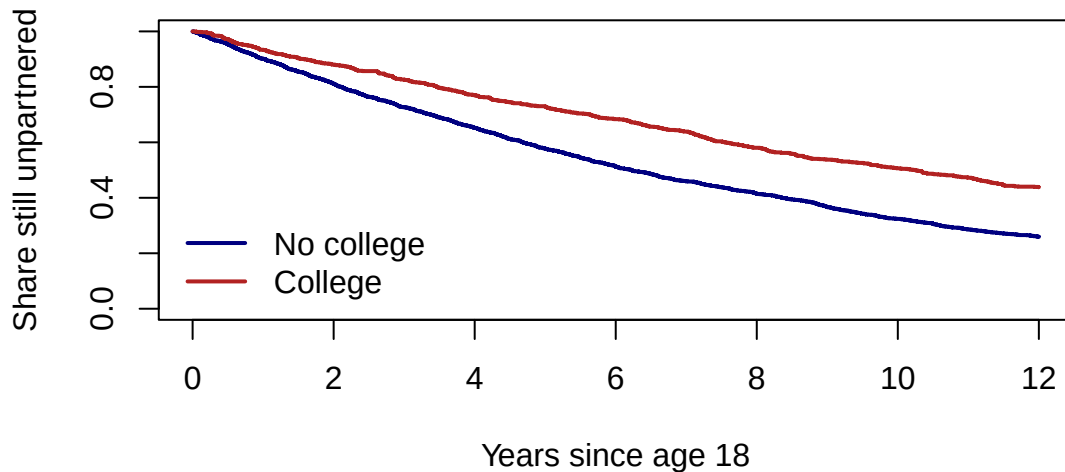


Figure 4: Kaplan–Meier survival curves by college. Each downward step is an observed union; censored units leave the risk set without a step.

8.4 The Cox model recovers the truth

```
fit_cox <- coxph(Surv(time, event) ~ college, data = sim)
summary(fit_cox)$coefficients |> round(3)
#>      coef exp(coef) se(coef)      z Pr(>|z|)
#> college -0.497    0.608   0.049 -10.09      0
```

The coefficient is the true -0.5 up to sampling noise, and $e^{\hat{\beta}} \approx 0.61$: college graduates form unions at about 61% of the non-college rate *at every point in time*. Censoring was handled by the likelihood, not by us.

8.5 Discrete time: the person-period logit

Cut time into years, give every person one row per year at risk, and run the logit from this week:

```
pp <- survSplit(Surv(time, event) ~ college, data = sim,
               cut = 1:11, episode = "period")
c(rows_people = n, rows_person_periods = nrow(pp))
#>      rows_people rows_person_periods
#>           3000           22661

fit_dt <- glm(event ~ college + factor(period),
              family = binomial, data = pp)
round(coef(fit_dt)["college"], 3)
#> college
#> -0.523
```

The person-period logit coefficient is essentially the Cox estimate: with small per-period hazards, the discrete-time logit and the proportional-hazards model agree. The period dummies play the

role of the baseline hazard, and censored people simply stop contributing rows — the likelihood logic from lecture, implemented by data construction.

i Note

Everything from earlier in this lab now applies to `fit_dt`: robust standard errors (cluster on the *person* — one person contributes many rows), average marginal effects, predicted per-period probabilities. That is the appeal of the discrete-time approach for sociologists.

9 Exercises

1. **Logit versus LPM for a causal contrast.** Estimate the “effect” of being ever-married (`evermar`) on holding a BA, controlling for `pareduc`, `wordsum`, `female`, `black`. Compare the LPM coefficient with the logit AME. How close are they? Which would you report, and to whom?
2. **The rescaling problem.** Fit the logit for `ba` with (a) `pareduc` only and
(b) `pareduc + wordsum + female + black`. Compare the `pareduc` *coefficient* across the two, then compare the `pareduc` *AME*. Explain the difference using Mood (2010).
3. **Simulation-based intervals.** Take the logit model and get an interval for the difference in predicted probability between `wordsum = 3` and `wordsum = 9` by simulating 10,000 draws of $\tilde{\beta}$ from $\mathcal{N}(\hat{\beta}, \hat{V}(\hat{\beta}))$ (King, Tomz, and Wittenberg 2000). Compare to the delta-method interval from `avg_comparisons()`.
4. **Offsets.** Suppose you wanted to model children *per year since age 18* rather than children. Add `offset(log(pmax(age - 18, 1)))` to the Poisson and interpret the coefficients. What changed?
5. **Zero inflation.** Fit `pscl::zeroinfl(childs ~ ... | evermar + age)` and compare with the negative binomial by AIC and by the predictive check above. Does the zero-inflation story make substantive sense here?
6. **Separation in the wild.** Fit a logit for `ba` including `factor(region) * black`. Does anything blow up? If so, apply Firth and report what changes.
7. **Censoring harder.** Rerun the event-history simulation with observation ending at 6 years instead of 12. How much censoring do you get now? Do Kaplan–Meier and the Cox model still recover the truth? What happens to the two naive estimates? (The lesson: the likelihood methods degrade gracefully; the naive ones get *worse* as censoring grows.)

10 Session information

```
sessionInfo()
#> R version 4.5.2 (2025-10-31)
```

```

#> Platform: aarch64-apple-darwin20
#> Running under: macOS Tahoe 26.6.2
#>
#> Matrix products: default
#> BLAS: /System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/vecLib.framework/Versions/A/Libraries/libBLAS.dylib
#> LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib;
#>
#> locale:
#> [1] C.UTF-8/C.UTF-8/C.UTF-8/C/C.UTF-8/C.UTF-8
#>
#> time zone: America/New_York
#> tzcode source: internal
#>
#> attached base packages:
#> [1] splines stats4 stats graphics grDevices utils datasets
#> [8] methods base
#>
#> other attached packages:
#> [1] VGAM_1.1-14 pROC_1.19.1 brglm2_1.1.0
#> [4] AER_1.2-17 car_3.1-5 carData_3.0-6
#> [7] MASS_7.3-65 lmtest_0.9-40 zoo_1.8-15
#> [10] sandwich_3.1-1 marginaleffects_0.32.0 ggplot2_4.0.2
#> [13] dplyr_1.2.0 survival_3.8-3
#>
#> loaded via a namespace (and not attached):
#> [1] generics_0.1.4 enrichwith_0.5.0 lattice_0.22-7
#> [4] digest_0.6.39 magrittr_2.0.4 evaluate_1.0.5
#> [7] grid_4.5.2 RColorBrewer_1.1-3 nleqslv_3.3.7
#> [10] fastmap_1.2.0 jsonlite_2.0.0 Matrix_1.7-4
#> [13] nnet_7.3-20 backports_1.5.0 Formula_1.2-5
#> [16] tinytex_0.58 scales_1.4.0 numDeriv_2016.8-1.1
#> [19] abind_1.4-8 cli_3.6.5 rlang_1.1.7
#> [22] withr_3.0.2 yaml_2.3.12 otel_0.2.0
#> [25] tools_4.5.2 checkmate_2.3.4 vctrs_0.7.1
#> [28] R6_2.6.1 lifecycle_1.0.5 insight_1.4.6
#> [31] pkgconfig_2.0.3 pillar_1.11.1 gtable_0.3.6
#> [34] Rcpp_1.1.1 data.table_1.18.2.1 glue_1.8.0
#> [37] statmod_1.5.2 xfun_0.56 tibble_3.3.1
#> [40] tidyselect_1.2.1 knitr_1.51 farver_2.1.2
#> [43] htmltools_0.5.9 labeling_0.4.3 rmarkdown_2.30
#> [46] compiler_4.5.2 S7_0.2.1

```